

Clinical Reasoning for Alzheimer's Disease from EEG Data Using EEGPT

Abstract

Introduction

Alzheimer's disease is the most common form of dementia, and the diagnostic tools tend to be expensive, invasive, or only useful late in the disease. EEG is appealing as a complement because it is cheap, non-invasive, and shows a consistent Alzheimer's Disease fingerprint. It is more slow-wave activity in delta and theta, weaker alpha and beta, lower signal complexity, and disrupted brain-state dynamics.

This project tests whether EEGPT, a pre-trained EEG foundation model that turns raw signals into dense latent embeddings, can be combined with a small prediction head and a set of clinically motivated biomarkers to separate Alzheimer's Disease from cognitively normal subjects across several public datasets. We compare two versions of the head: one using EEGPT embeddings alone, and one that concatenates the embeddings with handcrafted biomarkers (spectral ratios, alpha-2 features, microstates, Peak Alpha Frequency, Sample Entropy, and Lempel-Ziv complexity). A small instruction-tuned language model sits at the end of the pipeline to turn the numeric outputs into a readable clinical summary.

Methodology

Data source

Three OpenNeuro EEG datasets were used in this project, including both resting-state and task-based EEG recordings related to Alzheimer's disease. The two resting-state datasets, ds004504 and ds006036, share the same 88 subjects: 36 Alzheimer's disease (AD), 23 frontotemporal dementia (FTD), and 29 cognitively normal (CN) subjects. Their biomarker results were combined by averaging the eyes-closed and eyes-open results per subject, ensuring each subject contributed exactly one row to the combined analysis.

The third dataset, ds005048, contains 40 Hz auditory entrainment EEG recordings. After excluding MCI subjects (n=6) and subjects with missing diagnostic labels (n=2), 27 subjects were used: 17 AD and 10 CN. This dataset was included to evaluate biomarker generalizability across different EEG recording conditions.

Dataset	Recording Condition	Subjects	Diagnostic Groups
ds004504	Eyes-closed resting EEG	88	AD, FTD, CN
ds006036	Eyes-open EEG	88	AD, FTD, CN
ds005048	40 Hz auditory entrainment	27	AD, CN

Table 1: Dataset summary

For ease of processing, we rename datasets ds004504, ds006036, and ds005048 as **ds001**, **ds002**, and **ds003** respectively.

Preprocessing pipeline

The preprocessing pipeline was implemented for use on the KOA HPC cluster, designed based on the preprocessing methods used to generate derivative EEG datasets for ds004504 and ds006036. The pipeline processes raw EEGLAB .set files and outputs per-subject biomarker CSV files.

The preprocessing steps were:

1. **Channel normalization**

Channel names were standardized to match the standard 10-20 EEG montage. Old electrode names were converted to modern equivalents (e.g., T3→T7, T4→T8, T5→P7, T6→P8). Non-EEG channels including EOG, EMG, ECG, and trigger channels were removed.

2. **Bandpass filtering**

A 0.5–45 Hz fourth-order Butterworth bandpass filter was applied to remove low-frequency drift and high-frequency noise while preserving the frequency bands used for biomarker extraction.

3. **Artifact Subspace Reconstruction (ASR)**

ASR was applied to reduce high-amplitude artifacts. The cutoff was dataset-specific: ds004504 used a cutoff of 17, while ds006036 and ds005048 used a cutoff of 15. The stricter cutoff for ds004504 reflects differences in artifact patterns between eyes-closed and eyes-open recordings, where the latter tends to contain more ocular and attention-related artifacts.

4. **ICA and ICLabel artifact rejection**

ICA was performed using extended Infomax with 19 components at the original sampling

rate of each dataset (ds004504: 500 Hz, ds006036: 500Hz, ds005048: 250 Hz). ICLabel was then used to identify and reject ocular and muscle artifact components with probability exceeding 0.5. A minimum of two components was always retained to ensure sufficient signal remained after rejection.

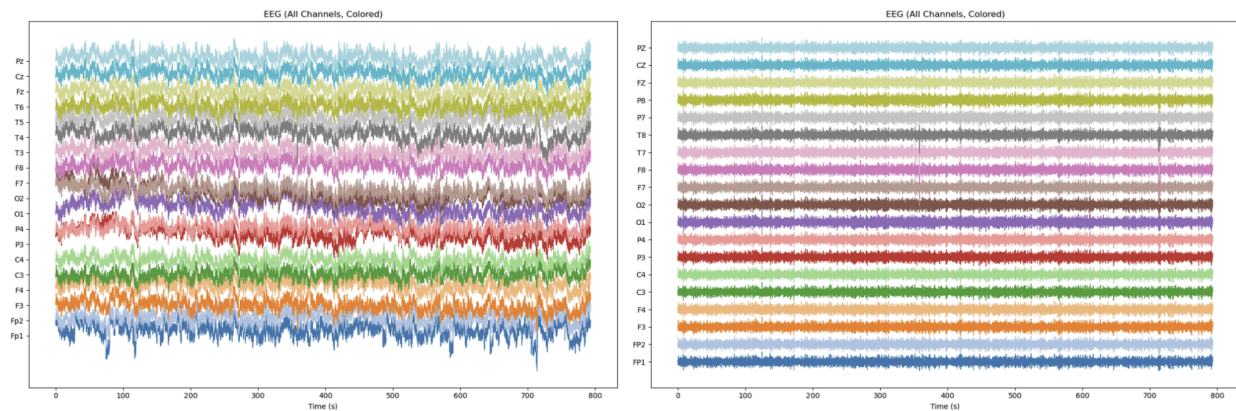


Figure 1: EEG signal comparison for sub-002 during eyes-closed recording. Left: before preprocessing. Right: after preprocessing.

Computing biomarkers / what biomarkers / what they mean

Biomarker 1: Spectral Power Ratios

The first biomarker category measures EEG slowing. Power spectral density was computed using Welch's method, and band powers were calculated for the delta, theta, alpha, beta, and gamma bands. EEG slowing is one of the most consistently reported quantitative EEG changes in Alzheimer's disease. Prior work reports that AD is associated with increased slow-frequency activity, particularly in the theta and delta bands, and reduced faster-frequency activity in the alpha and beta bands (Papaliagkas). This supports the use of theta/alpha ratio and DTABR as spectral biomarkers for dementia-related EEG slowing.

Biomarker	Meaning
Theta/Alpha Ratio	Ratio of theta to alpha power, indicating frequency-domain slowing
DTABR	Broad slowing index: $(\text{delta} + \text{theta}) / (\text{alpha} + \text{beta})$

Table 2: Spectral Power Ratio Biomarkers

Higher theta/alpha ratio and DTABR values indicate stronger dementia-like slowing. DTABR was used as a broader slowing index because it combines increased slow-wave activity with reduced faster activity, directly reflecting the reported AD pattern of increased delta/theta and reduced alpha/beta power (Papaliagkas). These features were used to evaluate whether AD and FTD subjects showed stronger slow-wave activity compared with cognitively normal subjects.

Biomarker 2: Alpha-2 Power and Complexity

The second biomarker category focuses on the alpha-2 band (10–13 Hz). Alpha-band activity is important because changes in alpha rhythms are consistently reported in AD and MCI. Prior studies have examined the relationship between upper and lower alpha activity using the alpha3/alpha2 ratio. Higher alpha3/alpha2 ratio has been associated with MCI subjects at higher risk of developing AD, as well as cortical thinning, hippocampal atrophy, and lower cerebral perfusion (Moretti, "Conversion"). A related study also discusses the alpha3/alpha2 power ratio as a potential EEG marker of AD-related cognitive decline (Patchitt et al.). Alpha-2 features were therefore included to examine whether changes in the upper alpha range could reflect AD-related disruption in resting-state EEG activity.

Biomarker	Meaning
Alpha2 Absolute Power	Absolute alpha-2 band power (V^2)
Alpha2 Relative Power	Alpha-2 power normalized by total band power
Alpha2 Spectral Entropy	Spectral complexity of the alpha-2 band
Alpha2 Peak Frequency	Dominant frequency within the alpha-2 range
Alpha2 Asymmetry	Left-right hemispheric alpha-2 power difference

Table 3: Alpha-2 Power and Complexity Biomarkers

Alpha-2 relative power was treated as a more stable feature than absolute power because it normalizes alpha-2 activity by total band power, partially reducing variability due to individual differences in recording conditions and anatomy. Alpha-2 spectral entropy was included because AD and MCI are associated with changes in EEG complexity. Prior qEEG work reports that EEG biomarkers for AD include altered rhythmic activity and reduced complexity (Papaliagkas). Spectral entropy was therefore used to examine whether the alpha-2 spectrum became more spectrally concentrated, indicating reduced complexity, across diagnostic groups.

Biomarker 3: EEG Microstates

The third biomarker category measures EEG microstate dynamics. Microstates represent short, recurring spatial patterns in EEG activity that capture large-scale brain-state dynamics beyond frequency-band power. Prior work describes EEG microstates as topographical maps representing discrete brain network activations, and AD has been associated with global microstate disorganization (Yang et al.).

The pipeline used ModKMeans clustering with four states labeled A, B, C, and D, matched to canonical reference topographies.

Biomarker	Meaning
Microstate Coverage	Fraction of time spent in each state
Mean Duration	Average time spent in each microstate visit (ms)
Occurrence	Number of state visits per second
GEV	Global explained variance of the microstate solution
Transition Probability	Probability of switching between states

Table 4: EEG Microstate Biomarkers

These features were used to evaluate whether dementia-related changes affected large-scale EEG brain-state dynamics. Prior microstate research has shown that duration, occurrence, coverage, GEV, and transition probability can serve as microstate-based features for AD classification (Yang et al.). In a related AD/FTD/control study, temporal parameters of microstates B and D differentiated healthy controls from dementia groups, and ratio-normalized microstate parameters improved group discrimination and correlated with Mini-Mental State Examination scores (MMSE). Among these, microstate B coverage was selected as the primary microstate biomarker based on its reported association with dementia-related changes in prior literature (Chang).

Microstate features were included as complementary biomarkers to spectral power features. Spectral ratios measure slowing in frequency content, while microstate features measure how brain-state patterns change over time. Together, these biomarkers provide a broader characterization of dementia-related EEG changes.

Biomarker 4: Peak Alpha Frequency

The fourth biomarker category measures Peak Alpha Frequency (PAF). This frequency is within the brain's alpha wave range from 8-12 Hz where the brain signal is strongest. A healthy brain is usually at 10 Hz, while an Alzheimer's brain is usually at 8 Hz.

Biomarker 5: Sample Entropy

The fifth biomarker category focuses on Sample Entropy. This measures how unpredictable the brain signal is moment-to-moment. Higher values mean the signal is more random. A healthy brain is complex and harder to predict, while an Alzheimer's brain is more repetitive and easier to predict, showing lower entropy.

Biomarker 6: Lempel-Ziv Complexity

The sixth biomarker category is Lempel–Ziv Complexity (LZC). This measures how many unique patterns are in the brain signal. A healthy brain uses a rich vocabulary, while an Alzheimer's brain repeats the same patterns more often.

EEGPT Encoder Model

The pretrained EEG foundation model proposed in the EEGPT study was used as the primary feature encoder for this project. The encoder transforms preprocessed EEG recordings into dense latent embeddings that capture temporal, spectral, and spatial characteristics of neural activity. For each EEG epoch, embeddings of dimension 512 were extracted from the encoder and used as input features for downstream Alzheimer's disease classification tasks. The encoder itself was initially frozen, allowing only the downstream prediction head to be trained. This approach was chosen to evaluate the representational quality of the pretrained EEG embeddings without altering the learned feature space.

Encoder Weight Unfreezing

To investigate if task-specific adaptation to the Alzheimers Disease EEG data could improve performance, we conducted additional experiments in which the upper transformer layers of the encoder were partially unfrozen and fine-tuned on the train datasets. Layers 7 and 8 of the encoder were unfrozen while the lower layers remained frozen. This, however, resulted in unstable optimization behavior with a rapidly increasing validation loss and reduced generalization. These experiments were discontinued due to limited project time and set aside as topics of future research.

Prediction Head Architecture

A lightweight neural prediction head was trained on top of the EEGPT embeddings for downstream clinical prediction. The prediction module consisted of:

1. A classification branch for predicting subject class:
 - Alzheimer's Disease (AD)
 - Healthy Control (HC)
2. A biomarker regression branch for predicting clinically relevant EEG biomarkers computed from the preprocessed EEG recordings.

As described above, the biomarkers included measures associated with spectral slowing and EEG complexity, including:

- DTABR (Delta-Theta / Alpha-Beta Ratio)
- Theta-to-Alpha Ratio
- Alpha-band spectral features
- Alpha spectral entropy
- EEG microstate coverage statistics

The prediction head jointly optimized classification and biomarker estimation objectives using a multi-task learning framework. The total loss function was defined as:

$$L_{\text{total}} = L_{\text{classification}} + \lambda L_{\text{biomarker}}$$

Where:

- $L_{\text{classification}}$ is the cross-entropy loss for AD/HC classification,
- $L_{\text{biomarker}}$ is the Huber regression loss for biomarker prediction,
- and λ controls the weighting between the two objectives.
 - λ was set to 0.3. Our goal was to let the model prioritize diagnostic prediction but still incorporate physiological information from the EEG-derived biomarkers.

Where the inclusion of biomarker prediction was intended to encourage the latent embedding space to preserve physiologically meaningful EEG characteristics associated with Alzheimer's disease progression.

Biomarker-Augmented Embeddings

Two experimental configurations were evaluated:

1. EEGPT embeddings only
2. EEGPT embeddings concatenated with handcrafted EEG biomarkers

This comparison was designed to determine whether explicitly incorporating domain-specific EEG biomarkers would improve predictive performance beyond the learned foundation model representations alone.

Large Language Model

To provide interpretable clinical explanations for model predictions, the Alibaba Cloud Qwen2.5-3B-Instruct model was integrated into the system as a lightweight clinical reasoning and summarization module.

The language model was prompted using:

- predicted diagnostic probabilities,
- EEG biomarker values,
- and model confidence scores

to generate natural-language clinical summaries describing potential neurological patterns associated with Alzheimer’s disease.

The Qwen2.5-3B-Instruct model was selected because it provides:

- strong instruction-following performance,
- relatively low computational requirements,
- and efficient inference on available hardware resources,

while remaining sufficiently capable for biomedical summarization and explanation tasks.

The language model was not used for diagnosis directly and instead served as an interpretability layer to translate quantitative model outputs into more human-readable clinical descriptions.

Experiments

Key Biomarkers

The key biomarkers identified from the combined resting-state dataset were:

Biomarker	Observed Pattern	Interpretation
DTABR	FTD $\approx$ AD > CN 31.2 $\approx$ 30.1 > 23.1	AD and FTD showed stronger broad EEG slowing than CN
Theta/Alpha Ratio	AD > FTD > CN 2.95 > 2.69 > 1.99	AD showed the strongest focused slowing relative to alpha
Alpha2 Absolute Power	CN > AD > FTD 1.82e-11 > 7.94e-12 > 3.48e-12	CN preserved stronger alpha-2 activity
Alpha2 Relative Power	CN > FTD $\approx$ AD 0.0376 > 0.0284 $\approx$ 0.0247	CN had more stable alpha-2 rhythm after normalization

Alpha2 Spectral Entropy	AD > FTD > CN 2.747 > 2.714 > 2.649	AD/FTD alpha-2 activity appeared less organized
Microstate B Coverage	FTD > AD > CN 0.257 ≈ 0.256 > 0.236	Dementia groups spent more time in microstate B

Table 5: Key EEG Biomarkers and Observed Patterns

The extracted biomarkers showed consistent group-level differences between cognitively normal subjects and dementia groups. In the combined resting-state dataset, theta/alpha ratio was highest in AD, followed by FTD and CN. DTABR was also higher in AD and FTD than CN, indicating stronger slow-wave activity in dementia groups. Alpha-2 relative power showed the opposite pattern, with CN having the highest value and AD/FTD showing lower values, suggesting reduced alpha-2 activity in dementia groups. Spectral entropy in the alpha-2 band was slightly higher in AD and FTD than CN, suggesting less organized alpha-band structure. Microstate B coverage was higher in FTD and AD than CN, indicating changes in large-scale EEG network dynamics. Together, these biomarkers provide converging evidence of EEG slowing, alpha rhythm degradation, and altered brain-state dynamics in dementia.

Synthetic EEG Data Generation (Cancelled Experiment)

Synthetic EEG data generation was implemented but not included in the final experiment. The goal was to augment the training data for EEGPT fine-tuning. Three augmentation methods were implemented:

- 1. Frequency augmentation**

FFT was applied to the EEG signal, and each frequency band amplitude was scaled by a random factor within $\pm 20\%$. IFFT was then used to reconstruct the signal. The phase was left unchanged to preserve synchronization.

- 2. Pink noise augmentation**

Pink noise ($1/f$ noise) was added at 20 dB SNR per channel. Pink noise was chosen because its spectral slope better matches the natural characteristics of EEG than white noise.

- 3. Combined augmentation**

Frequency augmentation and pink noise were applied together.

The synthetic data pipeline generated 12 copies per subject, producing 1,056 synthetic files per method (88 subjects $\times$ 12 copies). Metadata including group, age, and Mini-Mental State Examination (MMSE) score were inherited from participants.tsv based on subject ID. However, this experiment was excluded from the final analysis because there was insufficient time to validate whether synthetic EEG improved system performance. Since unvalidated synthetic data could introduce artificial patterns or bias into the model, exclusion was considered the more conservative approach.

UMAP Embedding Visualization

To investigate whether the pretrained EEGPT encoder produced meaningful latent representations for Alzheimer's disease classification, dimensionality reduction was applied to the extracted EEG embeddings using Uniform Manifold Approximation and Projection (UMAP).

Uniform Manifold Approximation and Projection is a nonlinear manifold learning technique commonly used for visualizing high-dimensional feature spaces in two dimensions while preserving local neighborhood structure. UMAP was selected because it provides more informative visualization of latent representation geometry compared to linear methods such as Principal Component Analysis (PCA).

For each experiment, the 512-dimensional EEGPT embeddings generated from the train and validation splits were projected into a shared two-dimensional latent space using the `umap-learn` implementation. The UMAP reducer was fit only on the training embeddings and subsequently used to transform the validation embeddings to ensure both splits occupied the same learned manifold space and to avoid information leakage from validation data during dimensionality reduction.

The following UMAP parameters were used:

- Number of components: $n=2$
- Number of neighbors: 15
- Minimum distance: 0.1
- Random seed: 42

The resulting 2D embeddings were visualized using scatter plots, with:

- red points representing Alzheimer's disease (AD) samples,
- and green points representing healthy control samples.

The purpose of this analysis was to qualitatively evaluate:

- class separability within the latent space,
- clustering behavior of EEG representations,
- and the degree to which the pretrained EEGPT embeddings naturally encoded disease-relevant structure prior to downstream classification training.

Distinct or partially separated clusters between AD and control samples would indicate that the encoder learned biologically meaningful representations associated with Alzheimer's disease-related EEG changes. Conversely, strong overlap between classes would suggest that additional fine-tuning or domain adaptation may be necessary to improve discriminative capability.

Training the Prediction Head

Dataset	Training Epochs	Validation Epochs	Training AD	Training Control	Validation AD	Validation Control
ds001	21,643	5,180	11,503	10,140	3,145	2,035
ds002	8,990	1,759	4,415	4,575	777	982
ds003	5,988	1,110	3,876	2,112	642	468
Combined	36,621	8,049	19,794	16,827	4,564	3,485

Table 6: Dataset distributions.

Biomarker	Mean	Standard Deviation
DTABR	20.5951	17.5703
theta_alpha_ratio	1.8779	1.4732
alpha2_abs	0.1110	0.3205
alpha2_rel	0.1899	0.3654
alpha2_spectral_entropy	2.2825	0.8639
ms_B_coverage	0.2410	0.0817

Table 7: Biomarkers used in the prediction head regression.

Fine-tuning the EEGPT Encoder

To adapt the encoder to our task, we selectively unfroze the last 2 transformer blocks (blocks 6 and 7) plus the final LayerNorm, approximately 6.3M of the 25M encoder parameters, while keeping the earlier 6 blocks fully frozen. Freezing the early layers theoretically preserves the general EEG knowledge acquired during pretraining; unfreezing only the last layers allows the model to adapt those high-level representations toward AD-specific patterns (alpha slowing, complexity reduction) without overwriting the foundational knowledge. We note that this portion of experiments was not completed due to poor results and requires further investigation.

Clinical Reasoning with LLM

We generate a structured report for each subject following the patient classification and biomarker prediction, which is passed to a local LLM to reason over the findings as a clinician would review a case to produce a narrative diagnostic impression.

For each of the six biomarkers, the report contains two values where available:

Model-predicted: estimated by the regression head from the EEGPT embedding. Fast, requires no additional signal processing, and available for any new patient whose EEG has been run through the pipeline.

Traditionally computed; derived from classical signal processing on the cleaned raw EEG (Welch PSD, microstate clustering). This is the ground truth value and is treated as more reliable.

Importantly, the reference values used for interpretation are derived from our own dataset (157 subjects across three recording conditions) rather than published literature values. This was a deliberate choice for two reasons:

1. First, our preprocessing pipeline differs from those used in published studies — different filtering, referencing, and normalization affect the absolute scale of spectral features. A DTABR of 24 in our pipeline may be equivalent to a DTABR of 31 in a study using different preprocessing. Using our own means ensures the boundaries are calibrated to what our pipeline actually produces.
2. Second, our dataset spans three fundamentally different recording conditions (eyes-closed resting, eyes-open photic stimulation, 40 Hz auditory entrainment). Eyes-open and auditory recordings suppress posterior alpha power relative to eyes-closed, which shifts the absolute values of every alpha-band biomarker. Literature values are almost exclusively derived from eyes-closed resting EEG, making them a poor reference for our mixed-condition dataset.

Normal range boundaries

Each biomarker's normal boundary is defined as: $\text{boundary} = (\text{AD group mean} + \text{CN group mean}) / 2$. This places the threshold at the midpoint between the two groups, which is the decision boundary a linear classifier would use under equal variance and equal priors. The direction of abnormality is then determined by which group has the higher mean.

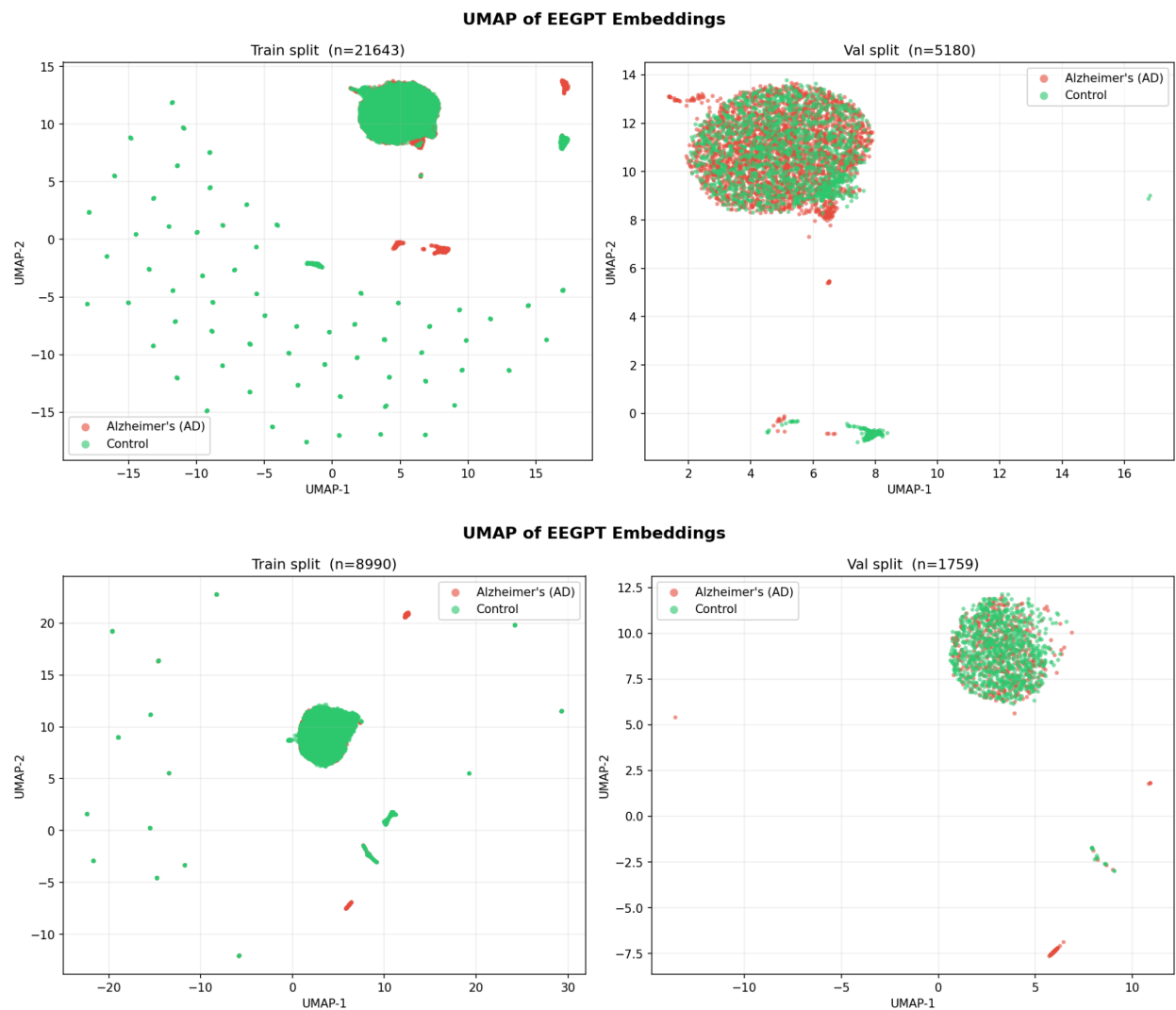
Biomarker	AD mean	CN mean	Boundary	Normal if
DTABR	24.15	18.42	21.28	≤ 21.28
theta_alpha_ratio	2.27	1.55	1.91	≤ 1.91
alpha2_abs	0.134	0.092	0.113	≥ 0.113

alpha2_rel	0.214	0.177	0.195	≥ 0.195
alpha2_spectral_entropy	2.474	2.276	2.375	≤ 2.375
ms_B_coverage	0.254	0.233	0.243	≤ 0.243

Table 8: Biomarker reference ranges for clinical reasoning.

Results

Embedding Space View



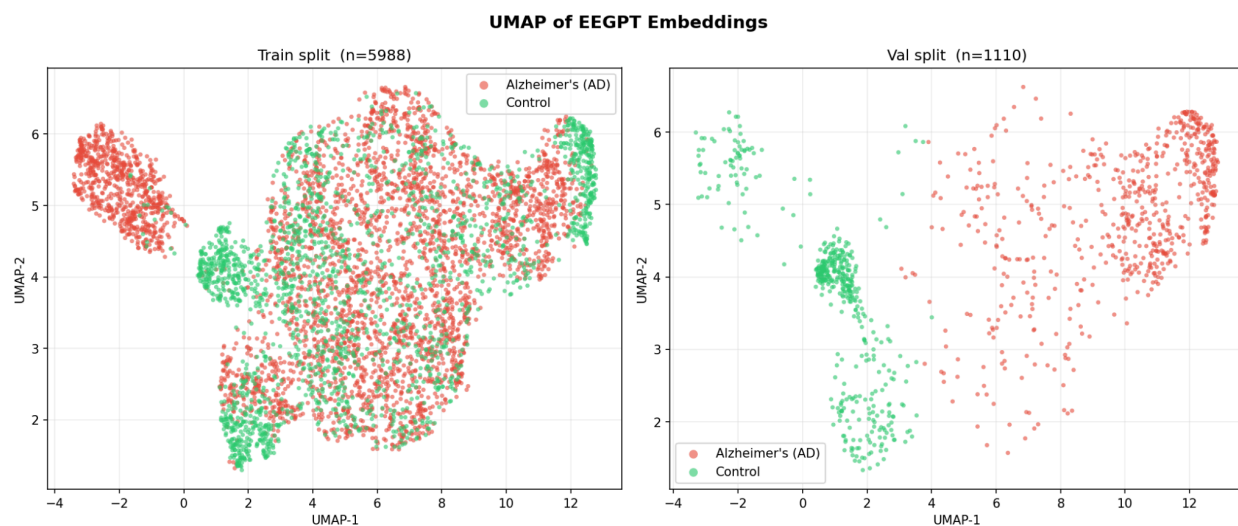


Figure 2: UMAP of EEGPT embeddings for the eyes-closed (top), eyes-open (middle), and auditory entrainment (bottom) datasets, split by train and validation set.

Validation Set Testing / Training

Datasets	Biomarkers	Best Val Loss	Best Val Acc	Best Val Bio	Early Stop Epoch	Head Params
ds001	No	0.8090	0.663	—	11	132,354
ds001 + ds002	No	0.6601	0.637	—	11	132,354
ds001 + ds003	No	0.6954	0.615	—	11	132,354
ds002 + ds003	No	0.6119	0.681	—	12	132,354
ds001 + ds002 + ds003	No	0.6489	0.663	—	11	132,354
ds001	Yes	0.6857	0.670	0.3661	11	133,896
ds001 + ds002	Yes	0.6639	0.631	0.2938	11	133,896
ds001 + ds003	Yes	0.6345	0.663	0.2149	11	133,896
ds002 + ds003	Yes	0.6208	0.627	0.1635	11	133,896

ds001 + ds002 + ds003	Yes	0.6175	0.656	0.2059	11	133,896
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Table 9: Comparison of combinations in dataset and biomarker inclusion in training.

Training Configuration and Dataset Composition

The EEGPT embeddings extracted from datasets ds001, ds002, and ds003 were combined into a single training corpus for prediction head training. All datasets used the same pretrained EEGPT encoder checkpoint, producing embeddings of dimension 512.

Metric	Value
Accuracy	64.2%
Balanced Accuracy	62.6%
AUC-ROC	0.6995
Sensitivity (AD Recall)	51.1%
Specificity (CN Recall)	74.2%
Precision (AD)	60.2%
F1 Score (AD)	55.3%
Biomarker Loss	0.1663

Table 10: Evaluation metrics for best head training on all three datasets and 6 biomarkers.

Validation Set Results

The trained EEGPT-based classification system was evaluated independently on the validation subsets of datasets ds001, ds002, and ds003. Performance was assessed using accuracy, balanced accuracy, AUC-ROC, sensitivity, specificity, precision, and F1-score. Because Alzheimer’s detection is a clinically sensitive task with potential class imbalance, balanced accuracy and sensitivity were considered particularly important evaluation metrics.

Dataset	Subjects	Accuracy	Balanced Accuracy	AUC-ROC	Sensitivity (AD Recall)	Specificity (CN Recall)
ds001	12	75.0%	72.9%	0.8571	85.7%	60.0%

ds002	12	58.3%	55.7%	0.4571	71.4%	40.0%
ds003	5	60.0%	58.3%	0.5000	66.7%	50.0%

Table 11: Performances of the combined 6-biomarker model on the three validation datasets.

Biomarker Selection Comparison

To evaluate the effect of biomarker selection on model performance, two multi-task training configurations were compared:

1. **A reduced biomarker set consisting of six clinically motivated biomarkers**
2. **A full biomarker set containing all 57 computed EEG biomarkers**

The goal of this experiment was to determine whether restricting the auxiliary regression task to a smaller set of clinically relevant biomarkers would improve classification performance and representation learning.

Metric	6 Biomarkers	57 Biomarkers
Accuracy	65.9%	61.8%
Balanced Accuracy	64.5%	60.4%
AUC-ROC	0.7115	0.6887
Sensitivity	54.2%	49.7%
F1 Score	57.9%	53.0%

Table 12: Comparison of performance by models trained with 6 and 57 biomarkers.

Configuration	Biomarker Loss
6 biomarkers	0.1682
57 biomarkers	0.1812

Table 13: Comparison of biomarker loss by models trained with 6 and 57 biomarkers.

Clinical Reasoning

The generated clinical reasoning reports contain several major components. They distinguish between predicted biomarker values and computed biomarker values. The reports also included caveats to improve transparency by explicitly communicating uncertainty instead of presenting predictions as definitive diagnoses.

Section	Description
Patient Information	Subject metadata including age, sex, MMSE score, and dataset
Classification Summary	AD vs control prediction probabilities and confidence
Biomarker Analysis	Interpretation of EEG biomarkers relative to AD patterns
Biomarker Ranking	Relative importance or deviation of biomarkers
Clinical Impression	Overall probabilistic diagnostic summary
Recommendations	Suggested follow-up assessments and imaging studies
Caveats and Warnings	Reliability concerns and uncertainty explanations

Table 14: Components of the generated clinical reasoning report.

Metric	Value
Subjects evaluated	770 (311 Dementia, 459 Normal)
Accuracy	49.0%
Balanced Accuracy	56.0%
AUC-ROC	0.6934
Sensitivity (AD recall)	92.9%
Specificity (CN recall)	19.2%

Metric	Value
Precision (AD)	43.8%
NPV (CN)	80.0%
F1 Score (AD)	0.595

Table 15. Evaluation metrics of our model on the CAUEEG test dataset.

Discussion

Embedding Space UMAP

The UMAP visualizations suggest that the pretrained EEGPT embeddings encode meaningful neurological structure associated with Alzheimer’s disease, though the degree of class separability varies substantially across datasets.

Key observations include:

- ds001 showed heavy class overlap and weak separability,
- ds002 demonstrated stable but mixed representations,
- ds003 exhibited the clearest latent disease structure.

These findings closely mirrored the downstream classification experiments, where dataset combinations involving ds003 consistently achieved stronger validation performance.

The variability in latent space organization across datasets further supports the hypothesis that dataset heterogeneity and domain shift represent major challenges for EEG foundation model generalization. Differences in EEG acquisition protocols, preprocessing pipelines, patient demographics, and diagnostic labeling criteria likely influence the learned embedding space and contribute to inconsistent classification behavior across datasets.

Overall, the UMAP analysis provides qualitative evidence that the EEGPT encoder learns partially discriminative representations of Alzheimer’s-related EEG activity prior to downstream supervised training, supporting the feasibility of foundation-model-based EEG representation learning for neurological disease analysis.

Training

The results in Table x indicate that the strongest signal comes from the combination of datasets, not biomarkers. This supports our hypothesis that the pretrained EEG embeddings do encode a large amount of discriminative information. It also suggests that biomarkers help more when the embedding space is weaker and less diverse. We see this, for instance, when we add the

biomarkers to ds001 alone, leading to a 0.1233 improvement in accuracy. However, it was still important for us to include those features in the model as they are important factors in the clinical reasoning process.

ds002 (eyes-open) + ds003 (auditory) was the best-performing combination of datasets, which indicates that ds001 might introduce a distribution shift. ds001 indeed appears to be the noisiest dataset, seen because adding any second dataset dramatically improves the ds001 loss.

We found that the model overfits extremely quickly, with early stopping around epoch 11-13. While the training loss steadily improves, validation loss worsens after the first or second epoch, indicating that the head is rapidly memorizing training examples then losing generalization. Because the head is so small at around 132k parameters with over 50% accuracy and fast convergence, this implies that the representation space is informative and demonstrates that the EEG encoder is indeed working.

Pertaining to the tests on the validation sets, the strongest performance was observed on ds001, where the model achieved 75.0% classification accuracy, 72.9% balanced accuracy, 0.8571 AUC-ROC. The model also demonstrated a relatively high sensitivity of **85.7%**, indicating that most Alzheimer's disease cases were successfully identified. Clinically, high sensitivity is valuable because false negatives in Alzheimer's screening may delay diagnosis and treatment.

Despite the small sample size, the relatively high AUC-ROC suggests that the learned EEGPT embeddings contained meaningful latent structure related to Alzheimer's disease patterns within this dataset.

However, performance on ds002 declined noticeably, with 58.3% accuracy, and a AUC-ROC below 0.5, indicating that the classifier's ranking behavior was effectively near-random. The results were similar for ds003, but the interpretation is limited due to the extremely small sample size of only five subjects. This suggests that the latent embedding space generated by EEGPT may not generalize consistently across datasets. Something to note is that ds001 was the largest dataset, which could have skewed the training.

As for cross-dataset trends, we noticed that the model consistently favored AD detection, with higher sensitivity than specificity, indicating the classifier was more effective in identifying Alzheimer's disease cases than healthy controls. In a clinical perspective, sensitivity can be more important because missing true Alzheimer's cases has greater consequences than false positives. The generalization across datasets being inconsistent again may indicate dataset-specific EEG characteristics, domain shift, and variability in recording EEG.

Biomarker Selection Comparison

The model trained using the reduced six-biomarker configuration outperformed the model trained using all 57 biomarkers across nearly all evaluation metrics. The reduced biomarker

configuration also achieved lower biomarker regression loss. This suggests that the smaller biomarker set was easier for the model to learn and may have provided cleaner supervisory signals during multi-task training. Carefully selected EEG biomarkers may better guide latent representation learning than a large collection of heterogeneous features. The all-57-biomarker model likely suffered from increased noise, feature redundancy, and more. Further exploration can most certainly be done into the best combination of biomarkers to use for both model performance and clinical reasoning.

Clinical Reasoning with LLM

One major limitation of deep learning systems in healthcare is their lack of interpretability. EEG embeddings generated by transformer encoders are highly abstract and difficult for clinicians to understand directly.

The clinical reasoning component addressed this limitation by translating latent model outputs into clinically meaningful explanations. Rather than providing only a binary prediction, the system produced biomarker-level evidence, confidence estimates, and textual summaries aligned with known Alzheimer's EEG findings. In future clinical workflows, such systems could potentially help neurologists rapidly review large EEG datasets and highlight patterns associated with neurodegenerative disease.

Limitations

A major limitation of this project was the heterogeneity of Alzheimer's disease EEG. Alzheimer's disease does not present a single uniform EEG profile. Subjects may differ in disease stage, severity, medication status, age, recording quality, and biological EEG characteristics. Because of this, even subjects within the same diagnostic group can show large differences in alpha power, theta activity, and microstate behavior. This issue was especially visible in alpha-2 absolute power. Absolute power values can vary strongly across individuals due to genuine biological differences and recording conditions such as electrode impedance, skull thickness, and hair. Therefore, a high standard deviation in alpha-2 absolute power does not indicate that the biomarker is invalid, but rather reflects real inter-subject variability. In contrast, alpha-2 relative power was more stable because it normalized alpha-2 power by total band power, reducing the effect of overall signal amplitude differences.

Task condition differences also affected biomarker comparison across datasets. The auditory dataset, ds005048, uses 40 Hz auditory entrainment, where the auditory stimulus entrains gamma-band activity that dominates the EEG spectrum. As a result, spectral biomarkers such as DTABR and theta/alpha ratio are not directly comparable to resting-state EEG values. For

this reason, the auditory dataset was interpreted separately as an independent validation of the pipeline under a different recording condition rather than merged with the resting-state datasets.

Generated reports are not medical diagnoses. The language model does not possess true clinical reasoning ability and instead generates explanations based on learned language patterns and structured model outputs. Consequently, all generated impressions still require confirmation through: physician review and other testing.

The reasoning system was trained and evaluated on relatively small EEG datasets with limited demographic diversity. As a result, biomarker thresholds may not generalize across institutions, EEG acquisition differences may affect interpretation quality, and generated reports may inherit biases present in the training data. Additionally, several biomarkers demonstrated substantial within-group variance, which reduces how reliable they are as features to use for supporting diagnoses.

Test Dataset

The model achieved an AUC-ROC of 0.694 on the CAUEEG test set, indicating above-chance discriminative ability between dementia and normal EEG signatures despite never having been trained or validated on this dataset. This result is notable given the substantial domain shift between training and test conditions: the model was trained on a combination of eyes-closed resting EEG (ds001), eyes-open photic stimulation EEG (ds002), and 40 Hz auditory entrainment EEG (ds003), while CAUEEG recordings consist of a mixed clinical protocol incorporating eyes-open, eyes-closed, and photic stimulation segments recorded in a Korean hospital setting at 200 Hz, subsequently resampled to 256 Hz for compatibility with the EEGPT encoder.

The most clinically salient finding is the model's high sensitivity of 92.9%, meaning it correctly identified 93 of every 100 true dementia cases. In a clinical screening context, high sensitivity is generally prioritized over specificity — missing a dementia case (false negative) carries greater clinical cost than a false alarm (false positive) that triggers further confirmatory testing. The trade-off, however, is a specificity of only 19.2%, indicating that the model classified the majority of normal subjects as having dementia. This sensitivity-specificity asymmetry suggests the model is operating with a decision boundary biased toward the positive (AD) class, likely because the training data had a higher proportion of AD epochs relative to controls, and the class-weighted loss function further amplified this bias.

Raw accuracy of 49.0% — below chance — is a direct consequence of this bias and should not be interpreted as the model performing worse than random. The dataset contains more Normal subjects (459) than Dementia (311), so a model that predicts everyone as Dementia would achieve approximately 40% accuracy while maximizing sensitivity. The balanced accuracy of 56.0% accounts for this imbalance and more accurately reflects the model's actual discriminative performance. The AUC-ROC of 0.694, which is threshold-independent, provides the most reliable summary of discriminative ability and confirms that the model has learned a meaningful signal.

Several factors likely contribute to the performance gap relative to training validation metrics. First, CAUEEG represents a distinct patient population — Korean clinical patients with a broader age range and different comorbidity profiles than the European research cohort used for training. Second, the recording protocol differences mean that the spectral characteristics of the EEG differ systematically: a mixed-condition recording with frequent task transitions produces different alpha and theta dynamics than a sustained resting-state recording. Third, the EEGPT encoder was pretrained on large general EEG datasets that may not have included clinical Korean EEG, meaning the embedding space for CAUEEG recordings may lie in a different region than the training data used to calibrate the prediction head. Finally, ICA-based artifact rejection was not applied during CAUEEG preprocessing due to computational constraints, meaning the encoder received noisier input than during training, potentially degrading embedding quality.

The model's behavior on MCI subjects — who were excluded from classification metrics but processed by the model — provides an additional qualitative validation. MCI is considered a prodromal stage of Alzheimer's disease, and a well-calibrated model should assign MCI subjects intermediate P(AD) values between those of Normal and Dementia subjects. Examining the distribution of predicted P(AD) scores across MCI subjects would reveal whether the model's learned representations capture this clinical gradient, which would constitute meaningful external validation of the model's neurological validity beyond simple accuracy metrics.

Overall, these results demonstrate that EEGPT-based transfer learning can generalize across datasets and recording conditions with clinically meaningful sensitivity, while highlighting the importance of domain-matched training data and preprocessing consistency for achieving balanced performance. Future work should explore fine-tuning on CAUEEG training data, post-hoc threshold calibration to balance sensitivity and specificity for the target clinical use case, and the inclusion of MCI as an intermediate class to more faithfully reflect the clinical spectrum of cognitive decline.

Future Work

A good deal of exploration was left to future work. This includes a more in-depth exploration of the biomarker space. One idea that was brought up but unexplored was the computation of feature importance for these biomarkers, such that the model could better identify which EEG-derived features contribute most strongly to Alzheimer's disease prediction. Techniques such as SHAP values, permutation importance, attention visualization, or ablation studies could help determine whether specific biomarkers consistently drive classification performance across datasets. This would improve model interpretability, potentially reduce redundancy in the biomarker set, and provide additional clinical insight into which electrophysiological changes are most associated with neurodegenerative progression.

We also propose a deeper exploration of fine-tuning the EEGPT encoder to better encode traits of Alzheimer’s Disease in its embeddings, which should improve separability of groups in the embedding space and lead to higher performance of the model predictions.

The current pipeline computes band power as mean PSD within each frequency band, which treats all bands equally regardless of bandwidth. Trapezoidal integration is the physically correct method for estimating total power (V^2) and would make absolute power biomarkers such as alpha-2 absolute power and delta absolute power directly comparable to reference values from existing EEG biomarker studies. Future work should compare classification performance between the two methods, particularly for absolute power features excluded from the current analysis.

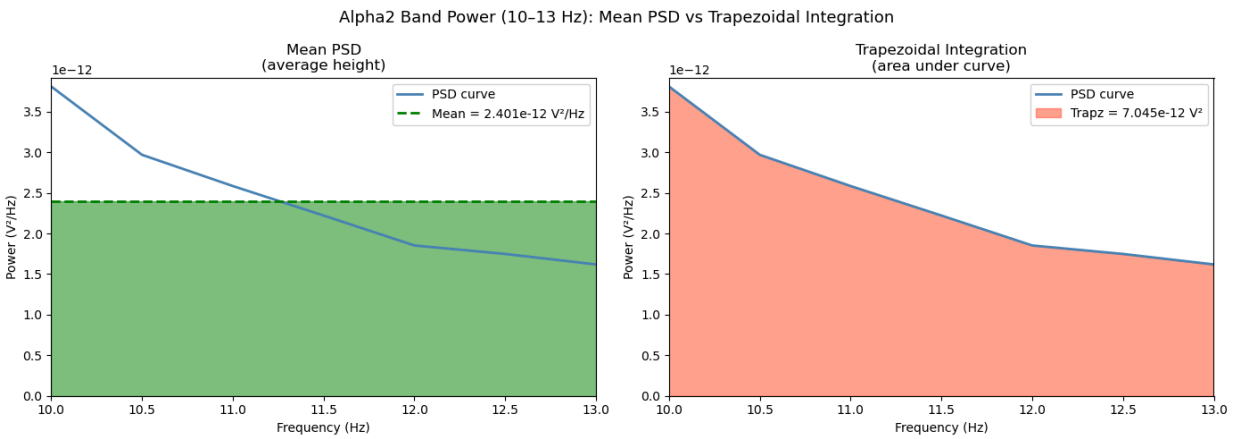


Figure 3: Mean PSD vs Trapezoidal Integration for Alpha2 band power (10–13 Hz). Trapz value is $\sim 3\times$ larger due to bandwidth scaling ($V^2/\text{Hz} \times 3 \text{ Hz} = V^2$).

Synthetic EEG generation was implemented but omitted from the final experiment due to time constraints. Future work should evaluate whether augmenting the training set with synthetic data improves classification performance, particularly for underrepresented groups such as FTD ($n=23$) where limited sample size may constrain model generalization.

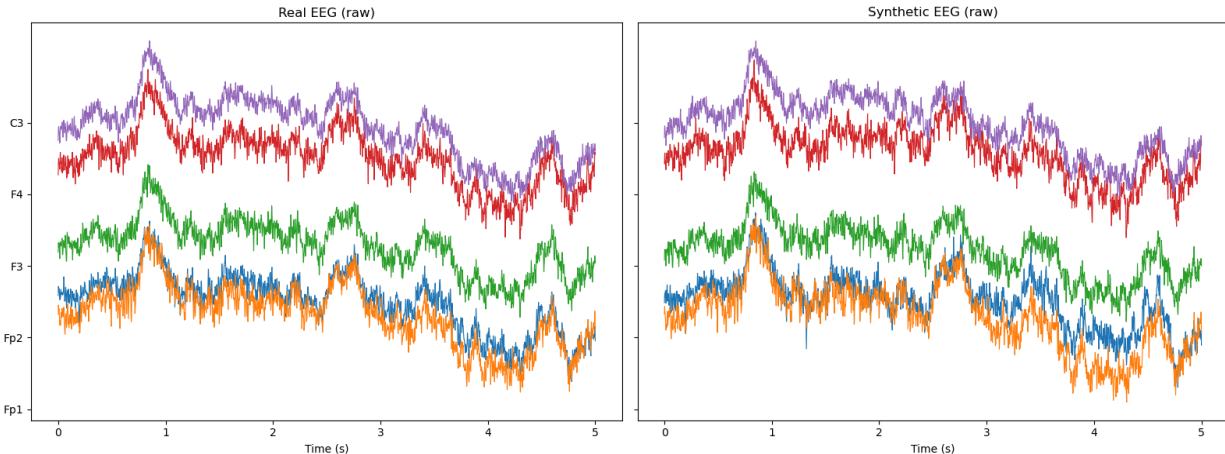


Figure 4: Real vs. synthetic EEG raw traces from the same subject (sub-002, AD)

Conclusion

The extracted biomarkers showed consistent group-level differences across multiple recording conditions, suggesting that the observed EEG patterns are not specific to a single dataset. In the combined resting-state dataset, theta/alpha ratio and DTABR were higher in AD and FTD than CN, confirming expected EEG slowing in dementia groups. Alpha-2 relative power was lower in AD and FTD, suggesting degraded upper alpha rhythm activity. Microstate B coverage was higher in dementia groups, indicating changes in large-scale brain-state dynamics. Together, these three biomarker categories provide converging evidence of EEG slowing, alpha rhythm degradation, and altered brain network dynamics associated with Alzheimer's disease, and were informative enough to support biomarker estimation in the pipeline and to provide interpretable evidence for clinical reasoning.

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Data Availability

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